

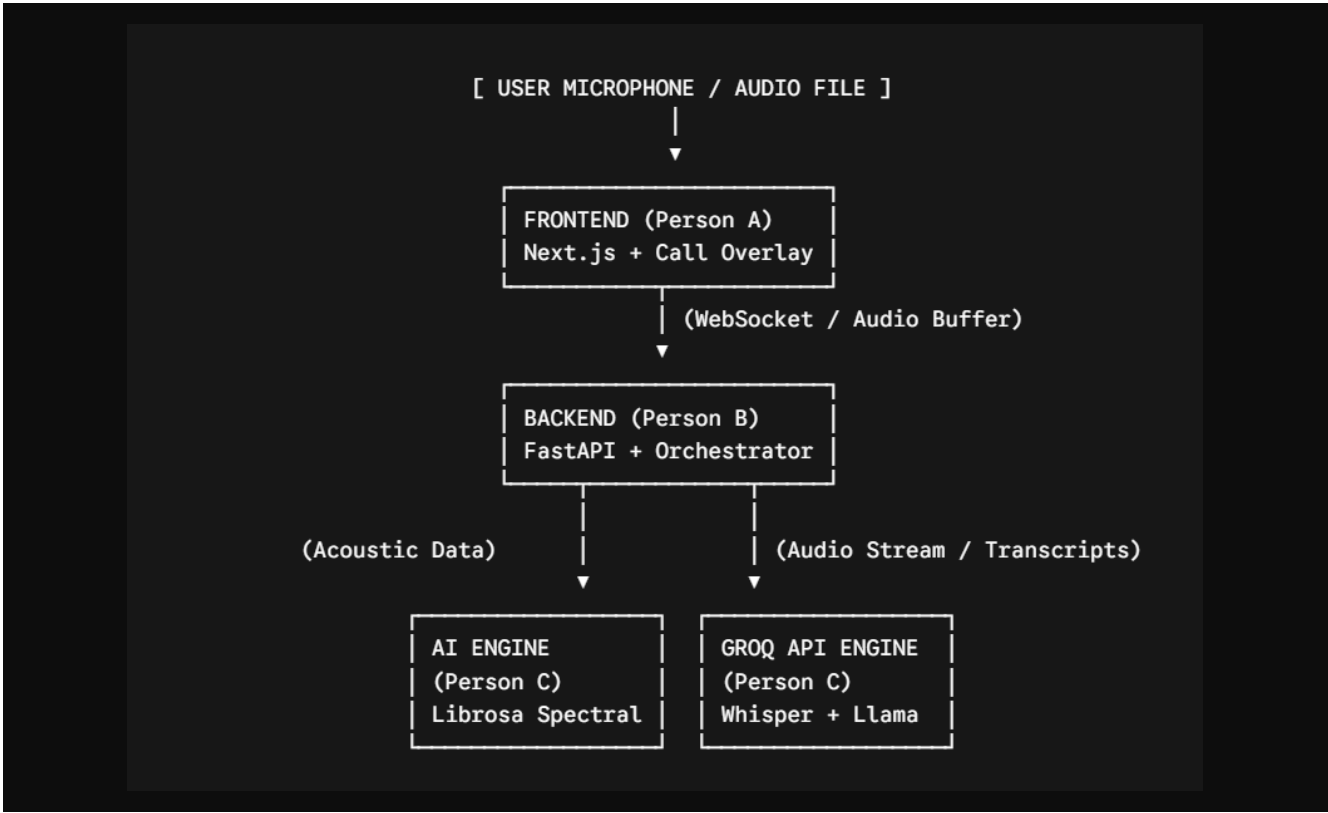
Tech Stack, Architecture & Work Division

VoiceLock AI — 36-Hour Build Plan for a 3-Person Team

Tech Stack & Architecture Map

Layer	Technology	Free Tier Details
Frontend UI	Next.js + Tailwind CSS	Deployed on Vercel (free continuous integration & custom domain).
Backend & WS	FastAPI (Python)	Deployed on Render.com or Koyeb (free web service hosting).
Speech-to-Text & LLM	Groq Cloud API	Free tier access to whisper-large-v3 & llama-3.1-8b-instant (ultra-fast, zero cost).
Acoustic AI	Python (librosa + scikit-learn)	Runs locally inside the Python backend process (zero external server cost).
Emergency Alerts	Telegram Bot API	100% free, instant SMS/push notification alternative sent directly to judges' phones.
Database & Logs	Supabase or Firebase	Free tier for storing call logs and threat history in real time.

System Architecture Flow



3-Person Work Division

■ Person A — Frontend & User Experience Lead

Primary Scope: Simulated phone call interface, audio capture, real-time threat gauge, and alert overlays.

- Build an authentic active phone call UI screen (looks like an incoming iOS/Android call).
- Integrate the Web Audio API to capture microphone audio in 500ms chunks (or allow uploading test audio files).
- Build a live visualizer (waveform) + Threat Meter Gauge (0% to 100%).
- Trigger full-screen red warning flashing states when the threat score exceeds 80%.
- Deploy the application to Vercel.

■ Person B — Backend Architect & Emergency Integrations

Primary Scope: FastAPI WebSocket server, threat fusion logic, database, and Telegram bot alert system.

- Set up a FastAPI WebSocket endpoint to ingest real-time audio streams from Person A.
- Receive analysis outputs from Person C (acoustic score + intent score) and compute the fused Threat Matrix Index.
- Integrate Telegram Bot API to send instant automated alerts ("CRITICAL ALERT: Voice Clone Scam Attempt Detected!") to emergency contacts.
- Set up Supabase/Firebase to save threat logs.
- Deploy the backend server to Render/Koyeb (with local tunneling fallback setup).

■ Person C — AI Engine & Audio Signal Processing Specialist

Primary Scope: Spectral vocoder artifact analysis, Groq API integration (Whisper + Llama 3.1).

- Acoustic Pipeline: write a Python module using librosa to compute MFCCs and spectral phase anomalies to detect synthetic audio signatures.
- Transcription Pipeline: pass audio buffers to Groq API's whisper-large-v3 for instant speech-to-text.
- Coercion NLP Engine: write a structured system prompt for Groq's llama-3.1-8b-instant to evaluate transcribed text for scam indicators ("bail," "wire money," "don't tell mom," "urgent transfer").
- Package AI responses into a clean JSON structure to pass to Person B.

36-Hour Master Sprint Schedule

Phase 1 · Foundation & Data Preparation (Hours 0–6)

Hour	Owner	Task
0–2	All	Project kickoff. Finalize JSON schema between FE, BE, and AI. Get free API keys for Groq, Supabase, and Telegram.
2–4	Person C	Collect 5 real human audio clips and generate 5 synthetic AI voice clips (ElevenLabs free tier / open tools) as testing data.
2–6	Person A	Scaffold Next.js project with Tailwind CSS; build basic incoming call screen template.
2–6	Person B	Scaffold FastAPI server, WebSocket endpoint, and connect Telegram Bot.

MILESTONE 1 (Hour 6): Basic FE connects to BE via WebSocket; API keys are verified working.

Phase 2 · Core Feature Build (Hours 6–18)

Hour	Owner	Task
6–12	Person A	Implement browser microphone recording (MediaRecorder API) and streaming via WebSockets. Build audio file upload toggle for demo fallback.
6–12	Person B	Build threat fusion scoring logic in FastAPI: $\text{Threat Score} = (0.5 \times \text{Acoustic Score}) + (0.5 \times \text{Coercion Score})$.
6–12	Person C	Finalize librosa feature extraction script and build Groq Llama prompt for structured JSON scoring.
12–18	Person A	Build high-contrast threat alert UI overlay and “Alert Emergency Contact” trigger button.
12–18	Person B	Connect Telegram Bot alert trigger when threat score > 80%. Connect Supabase logging.
12–18	Person C	Combine librosa output + Groq Whisper + Groq Llama into a single fast function returning JSON.

MILESTONE 2 (Hour 18): End-to-end pipeline works — audio input streams data, calculates risk score, and triggers a Telegram alert.

Phase 3 · Integration, Polish & Deployment (Hours 18–30)

Hour	Owner	Task
18–24	Person A	Deploy frontend to Vercel. Connect production WebSockets URL.
18–24	Person B	Deploy backend to Render.com (or set up ngrok tunnel as an immediate zero-latency backup).
18–24	Person C	Test edge cases (normal friendly conversation vs. aggressive fake emergency scam call) to fine-tune threat weights and eliminate false positives.
24–30	Person A	Add smooth gauge animations, glowing red radar pulses on high alerts, and live audio waveform visualizers.
24–30	Person B & C	Optimize API latency to ensure the whole pipeline executes in under 1 second per audio chunk.

MILESTONE 3 (Hour 30): Full MVP is live online, hosted on free tiers, and passes live test scenarios.

Phase 4 · Pitch Prep & Backup Safeguards (Hours 30–36)

Hour	Owner	Task
30–33	All	Perform 5 full end-to-end live trial runs.
33–34	Person A & B	CRITICAL: Record a crisp 30-second screen recording of the working MVP (screen + Telegram alert) as a Wi-Fi-failure backup for live judging.
34–36	Person C & Team	Prepare the 3-minute pitch deck emphasizing societal impact, zero-cloud privacy architecture, and real-time live performance.